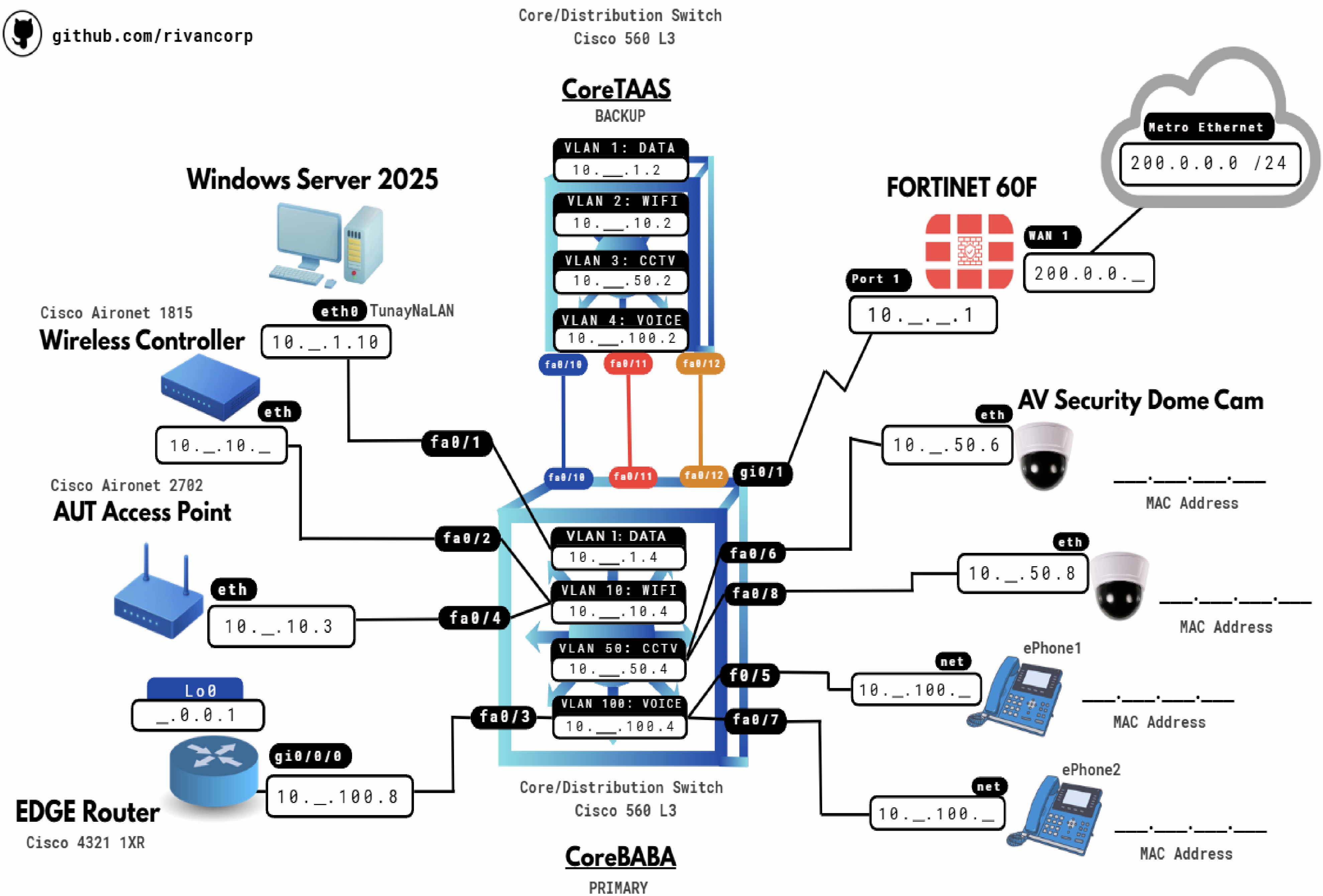
REAL WORLD LABS

RIVAN DAY 1 LABS

github.com/rivancorp



Master The CISCO Command Line Interface

CLI Command:

- **SR** = Show Run
- **SS** = Show Start
- **CRS** = Copy Run Start
- **SIIB** = Show Ip Int Brief
- **SVB** = Show Vlan Brief
- **SIT** = Sh Int Trunk
- **SMAC** = Sh Mac Address-Table
- **SIR** = Sh Ip Route

(yes/no?) — **SETUP / INUTIL Mode**

↑ ↓ ctrl + c

Router> — **USER / FRIENDZONE Mode**
Few Show, No Touch

disable ↑ ↓ enable

Router# — **PRIVILEGE / CUBAO Mode**
All Show, No Touch

exit end ctrl + c ↑ ↓ configure terminal

Router(config)# — **GLOBAL / PASIG Mode**
All Touch, No Show

exit ↑ ↓
hostname
enable secret pass
service password-encryption
no logging console
no ip domain lookup

LINE
ine console 0
ine vty 0 14
password pass
login
exec-timeout 0 0

INTERFACE
interface fa0/0
no shutdown

ROUTER
router ospf 1
router-id
network

DHCP
ip dhcp pool test
network
default-router
domain-name
dns-server
option 150 ip

TELEPHONY
telephony-service
no auto-assign
no auto-reg-ephone
max-ephones
max-dn
ip source address

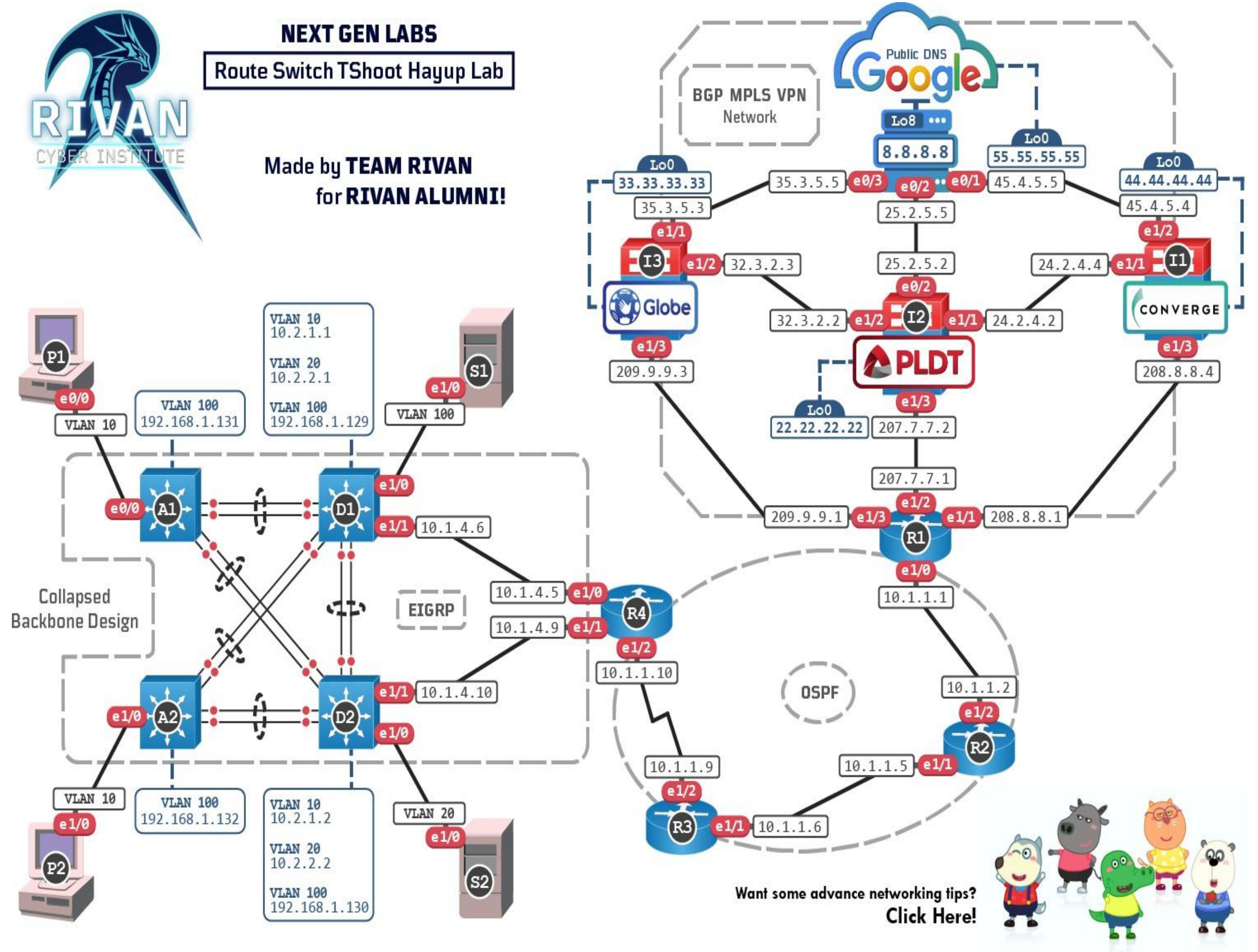
VOICE-PORT
voice-port 0/0/0
no shutdown

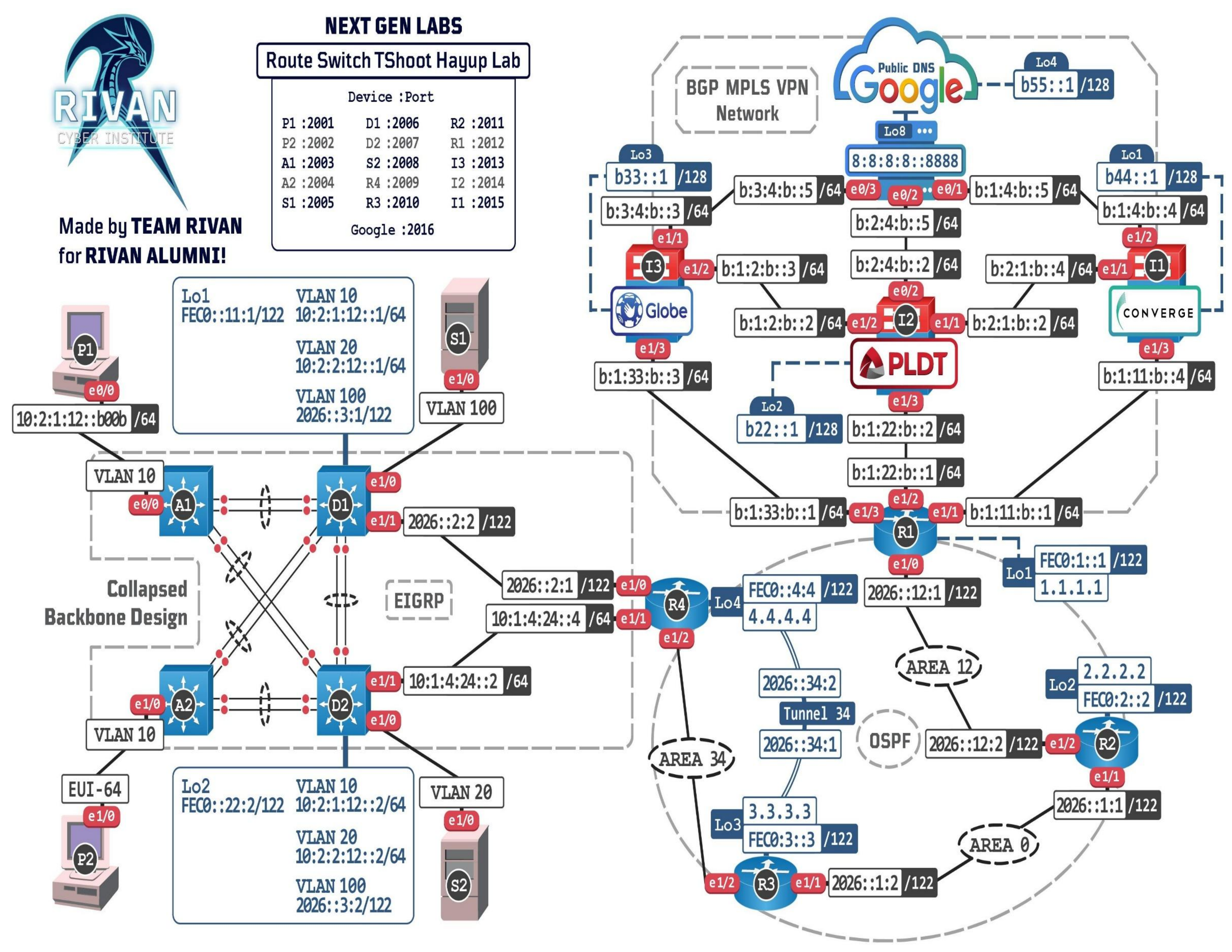
ACL
no access-list 1

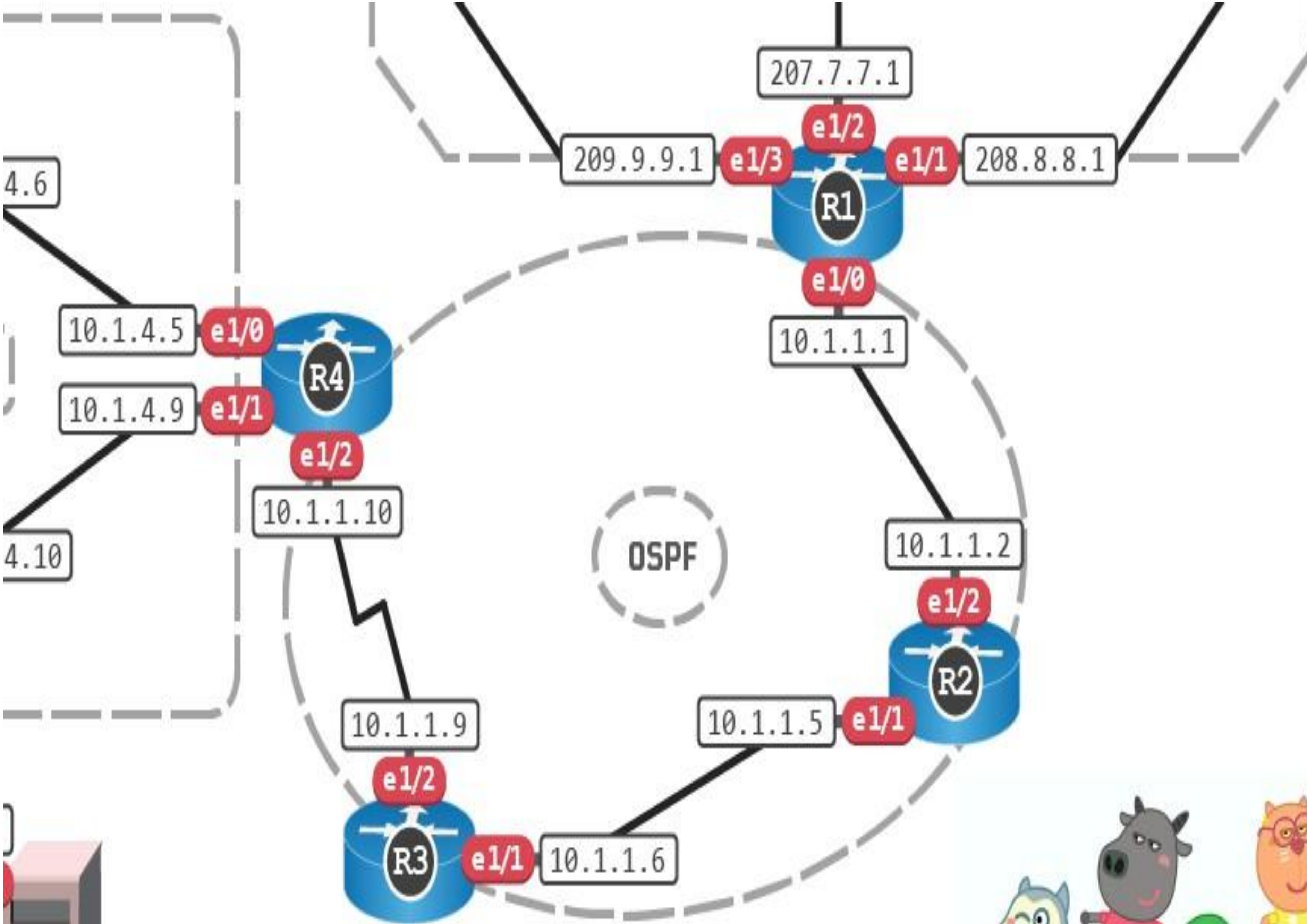
NEXT GEN LABS

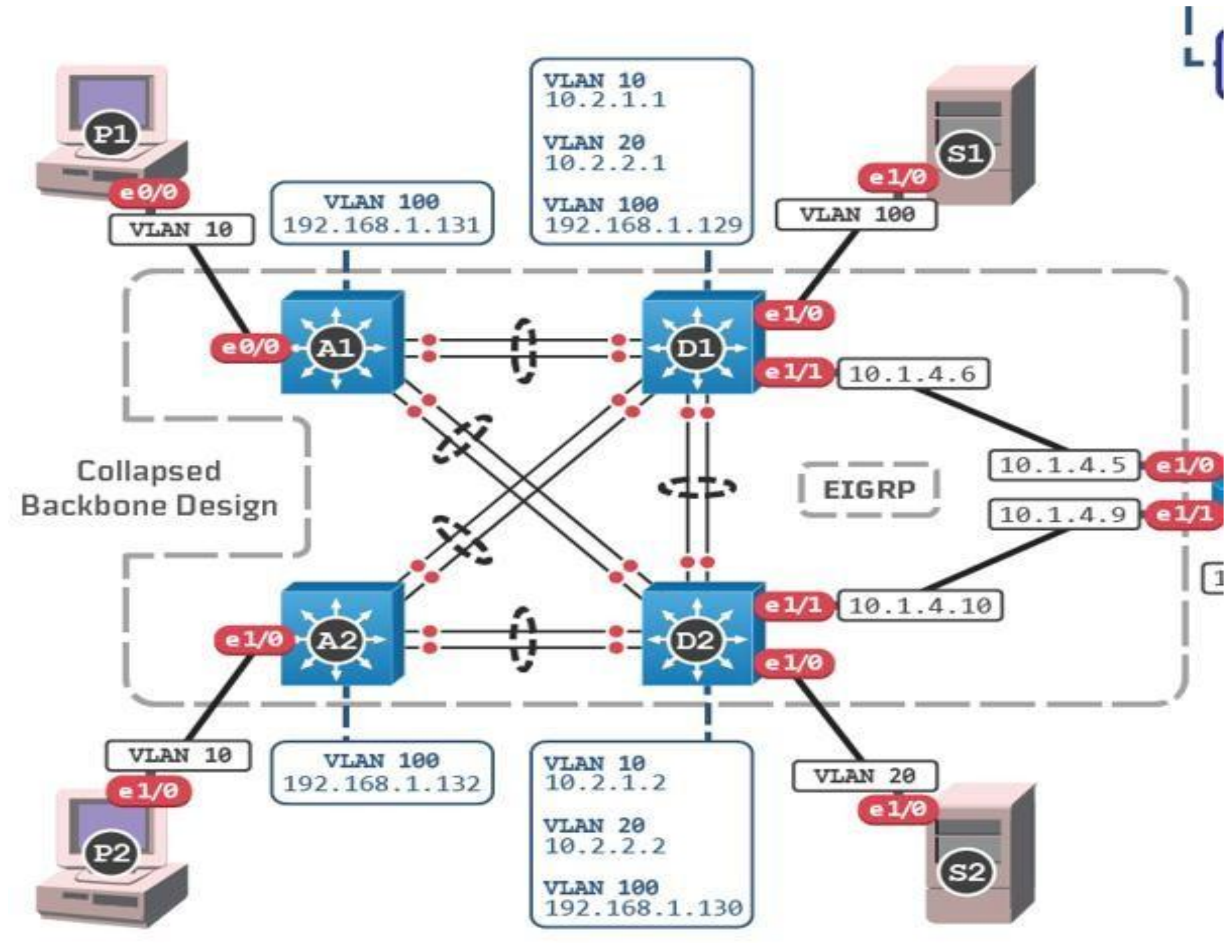
Route Switch TShoot Hayup Lab

Made by **TEAM RIVAN** for **RIVAN ALUMNI!**




RIVAN
CYBER INSTITUTE
Made by TEAM RIVAN
for RIVAN ALUMNI!
NEXT GEN LABS
Route Switch TShoot Hayup Lab
Device :Port
P1 :2001
P2 :2002
A1 :2003
A2 :2004
S1 :2005
D1 :2006
D2 :2007
S2 :2008
R4 :2009
R3 :2010
R2 :2011
R1 :2012
I3 :2013
I2 :2014
I1 :2015
Google :2016
Lo1
FEC0::11:1/122
VLAN 10
10:2:1:12::1/64
VLAN 20
10:2:2:12::1/64
VLAN 100
2026::3:1/122
Lo2
FEC0::22:2/122
VLAN 10
10:2:1:12::2/64
VLAN 20
10:2:2:12::2/64
VLAN 100
2026::3:2/122
P1
e0/0
10:2:1:12::b00b /64
VLAN 10
S1
e1/0
VLAN 100
A1
e0/0
D1
e1/0
e1/1
2026::2:2 /122
Collapsed
Backbone Design
EIGRP
A2
e1/0
VLAN 10
D2
e1/1
10:1:4:24::2 /64
e1/0
EUI-64
e1/0
P2
VLAN 20
e1/0
S2
2026::2:1 /122
10:1:4:24::4 /64
e1/0
e1/1
R4
e1/2
Lo4
FEC0::4:4 /122
4.4.4.4
2026::34:2
Tunnel 34
2026::34:1
AREA 34
Lo3
3.3.3.3
FEC0:3::3 /122
R3
e1/2
e1/1
2026::1:2 /122
AREA 0
2026::1:1 /122
e1/1
R2
e1/2
Lo2
2.2.2.2
FEC0:2::2 /122
2026::12:2 /122
OSPF
AREA 12
2026::12:1 /122
e1/0
R1
Lo1
FEC0:1::1 /122
1.1.1.1
e1/3
e1/2
e1/1
b:1:33:b::1 /64
b:1:22:b::1 /64
b:1:11:b::1 /64
BGP MPLS VPN
Network
Public DNS
Google
Lo4
b55::1 /128
Lo8
8:8:8:8::8888
e0/3
e0/2
e0/1
b:3:4:b::5 /64
b:1:4:b::5 /64
b:2:4:b::5 /64
b:2:4:b::2 /64
Lo3
b33::1 /128
b:3:4:b::3 /64
e1/1
I3
e1/2
b:1:2:b::3 /64
Globe
e1/3
b:1:33:b::3 /64
Lo1
b44::1 /128
b:1:4:b::4 /64
e1/2
I1
e1/1
b:2:1:b::4 /64
CONVERGE
e1/3
b:1:11:b::4 /64
e0/2
b:1:2:b::2 /64
e1/2
I2
e1/1
b:2:1:b::2 /64
PLDT
e1/3
Lo2
b22::1 /128
b:1:22:b::2 /64


207.7.7.1
e1/2
209.9.9.1
e1/3
e1/1
208.8.8.1
R1
e1/0
10.1.1.1
4.6
10.1.4.5
e1/0
R4
10.1.4.9
e1/1
e1/2
10.1.1.10
4.10
OSPF
10.1.1.2
e1/2
R2
10.1.1.5
e1/1
10.1.1.9
e1/2
R3
e1/1
10.1.1.6
S2



P1
e0/0
VLAN 10
VLAN 100
192.168.1.131
VLAN 10
10.2.1.1
VLAN 20
10.2.2.1
VLAN 100
192.168.1.129
S1
e1/0
VLAN 100
e0/0
A1
D1
e1/0
e1/1
10.1.4.6
Collapsed
Backbone Design
EIGRP
10.1.4.5
e1/0
10.1.4.9
e1/1
e1/0
A2
D2
e1/1
10.1.4.10
e1/0
VLAN 10
e1/0
P2
VLAN 100
192.168.1.132
VLAN 10
10.2.1.2
VLAN 20
10.2.2.2
VLAN 100
192.168.1.130
VLAN 20
e1/0
S2

OSI (Open Source Interconnection) 7 Layer Model

DEPARTMENT OF DEFENSE (USA)

#	LAYER		APPLICATION / EXAMPLE		CONTROL DEVICES / PROTOCOLS	DOD4 MODEL	
7	**APPLICATION LAYER**	Serves as the window for users and application processes to access the network services.	END USER LAYER	Program that opens what was sent or creates what is to be sent. Resource Sharing • Remote File Access • Remote Printer Access • Directory Services • Network Management	USER APPLICATIONS **Example:** Messenger, VLC, HTTP, HTTPS **SMTP** (Simple Mail Transfer Protocol)	PROCESS	APPLICATION
6	**PRESENTATION LAYER**	Formats the data to be presented to the Application layer. It can be viewed as "Translator" for the network. DATA FORMAT	SYNTAX LAYER	Encrypt and Decrypt (If Needed) CODEC / VLC | .mp4, .mkv, .m4a Character Code Translation • Data Conversion • Data Compression • Data Encryption • Character Set Translation  DATA	DATA FORMATS/BYTES FORMAT JPEG/ASCII,EBDIC/TIFF/GIF PICT .exe .bat .ps/.pl .vbs .py	PROCESS	APPLICATION
5	**SESSION LAYER**	The beginning of hacking session. HACKER --> VICTIM	SYNCH and SEND to PORTS	Data becomes small (Logical Ports) NET_USE \\ 10.10.10.10 \ IPC$ | NET STAT -s -p TCP Session Establishment, Maintenance, and Termination • Session Support - Perform Security, Name Recognition, Logging, etc.  DATA STREAM	LOGICAL PORTS RPC/SQL/NFS/IPC NetBIOS Names "Data Streams"	PROCESS	APPLICATION
4	**TRANSPORT LAYER** ♥ HACKER	Ensures that messages are delivered error-free, in sequence, and with no loses or duplications.	TCP / UDP	Host to Host (Flow Control) NMAP -v 10.10.10.10/24 Message Segmentation • Message Acknowledgement • Message Traffic Control • Session Multiplexing TCP vs UDP PACKET FILTERING	PROTOCOL DATA UNIT TCP/SPX/UDP TCP-*reliable* UDP-*unreliable* "Segments"	HOST TO HOST	TRANSPORT
3	**NETWORK LAYER** ♥ HACKER	Controls the operations of the subnet, deciding which physical path the data takes. ROUTER	IP PACKETS	("Letter", contains IP Address) NMAP -sP 10.10.10.10/24 Routing • Subnet Traffic Control • Frame Fragmentation • Logical-Physical Address Mapping • Subnet Usage Accounting PACKET FILTERING	ROUTERS IP/PX/ICMP "Packets"	INTERNET	INTERNET
2	**DATA LINK LAYER**	Provides error-free transfer of data frames from one node to another over the Physical layer. SWITCH and ACCESS POINT	FRAMES	("Envelopes", contains MAC Address) NIC card ---- Switch ---- NIC card (end to end) Establishes & Terminates the Logical Link between Nodes • Frame Traffic Control • Frame Sequencing • Frame Acknowledgement • Frame Delimiting • Frame Error Checking • Media Access Control ACCESS POINT	SWITCH BRIDGE WAP PPL/SLIP LAN BASED LAYERS	NETWORK	NETWORK ACCESS
1	**PHYSICAL LAYER**	Concerned with the transmission and reception of the unstructured raw bit stream over the physical medium.	PHYSICAL STRUCTURE	Cables, Hubs, etc. V - oltage -> (PoE - Power over Ethernet | 45v & 53v) I - nterface -> (L.C - Lucent Connector | RJ45) T - ransmission -> 1. Fiber Optic (850 Laser & 1015 Laser) 2. Radio Frequency (Wireless | 2.4 GHz & 5 GHz) 3. UTP (Electrical) Data Encoding • Physical Medium Attachment • Transmission Technique - Baseband or Broadband • Physical Medium Transmission Bits & Volts **BYTES** - measure data **BITS** - physical transmission	HUB Send and Receive "Bits" LAN BASED LAYERS	NETWORK	NETWORK ACCESS

GATEWAY

ENCAPSULATION